

Foundations of Momentum optimization

[Momentum optimization]

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$$v(w, t) := \gamma v(w, t-1) + (1-\gamma) (\nabla Q_i(w))^2 \quad \{\displaystyle v(w,t):=\gamma v(w,t-1)+\left(1-\gamma\right)\left(\nabla Q_{\{i\}}(w)\right)^2\}$$

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Averaging Averaged stochastic gradient descent, invented independently by Ruppert and Polyak in the late 1980s, is ordinary stochastic gradient descent that records an average of its parameter vector over time. That is, the update is the same as for ordinary stochastic gradient descent, but the algorithm also keeps track of

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History In 1951, Herbert Robbins and Sutton Monro introduced the earliest stochastic approximation methods, preceding stochastic gradient descent. Building on this work one year later, Jack Kiefer and Jacob Wolfowitz published an optimization algorithm very close to stochastic gradient descent, using central differences as an approximation of the gradient. Later in the 1950s, Frank Rosenblatt used SGD to optimize his perceptron model, demonstrating the first applicability of stochastic gradient descent to neural networks. Backpropagation was first described in 1986, with stochastic gradient descent being used to efficiently optimize parameters across neural networks with multiple hidden layers. Soon after, another improvement was developed: mini-batch gradient descent, where small batches of data are substituted for single samples. In 1997, the practical performance benefits from vectorization achievable with such small batches were first explored, paving the way for efficient optimization in machine learning. As of 2023, this mini-batch approach remains the norm for training neural networks, balancing the benefits of stochastic gradient descent with gradient descent. By the 1980s, momentum had already been introduced, and was added to SGD optimization techniques in 1986. However, these optimization techniques assumed constant hyperparameters, i.e. a fixed learning rate and momentum parameter. In the 2010s, adaptive approaches to applying SGD with a per-parameter learning rate were introduced with AdaGrad (for "Adaptive Gradient") in 2011 and RMSprop (for "Root Mean Square Propagation") in 2012. In 2014, Adam (for "Adaptive Moment Estimation") was published, applying the adaptive approaches of RMSprop to momentum; many improvements and branches of Adam were then developed such as Adadelta, Adagrad, AdamW, and Adamax. Within machine learning, approaches to optimization in 2023 are dominated by Adam-derived optimizers, TensorFlow and PyTorch, by far the most popular machine learning libraries, as of 2023 largely only include Adam-derived optimizers, as well as predecessors to Adam such as RMSprop and classic SGD. PyTorch also partially supports limited-memory BFGS, a line-search method, but only for single-device setups without parameter groups.

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where i $\{\displaystyle i\}$ is uniformly sampled between 1 and n $\{\displaystyle n\}$. Although theoretical convergence of this procedure happens under relatively mild assumptions, in practice the procedure can be quite unstable. In particular, when η $\{\displaystyle \eta\}$ is misspecified so that $I - \eta x_i x_i'$ $\{\displaystyle I-\eta x_{\{i\}}x_{\{i\}}'\}$ has large absolute eigenvalues with high probability, the procedure may diverge numerically within a few iterations. In contrast, implicit stochastic gradient descent (shortened as ISGD) can be solved in closed-form as:

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Nesterov-enhanced gradients: NAdam, FASFA varying interpretations of second-order information: Powerpropagation and AdaSqrt. Using infinity norm: AdaMax AMSGrad, which improves convergence over Adam by using maximum of past squared gradients instead of the exponential average. AdamX further improves convergence over AMSGrad. AdamW, which improves the weight decay.

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where the parameter w that minimizes $Q(w)$ is to be estimated. Each summand function Q_i is typically associated with the i -th observation in the data set (used for training). In classical statistics, sum-minimization problems arise in least squares and in maximum-likelihood estimation (for independent observations). The general class of estimators that arise as minimizers of sums are called M-estimators. However, in statistics, it has been long recognized that requiring even local minimization is too restrictive for some problems of maximum-likelihood estimation. Therefore, contemporary statistical theorists often consider stationary points of the likelihood function (or zeros of its derivative, the score function, and other estimating equations). The sum-minimization problem also arises for empirical risk minimization. There, $Q_i(w)$ is the value of the loss function at i -th example, and $Q(w)$ is the empirical risk. When used to minimize the above function, a standard (or "batch") gradient descent method would perform the following iterations:

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Notable applications Stochastic gradient descent is a popular algorithm for training a wide range of models in machine learning, including (linear) support vector machines, logistic regression (see, e.g., Vowpal Wabbit) and graphical models. When combined with the backpropagation algorithm, it is the de facto standard algorithm for training artificial neural networks. Its use has been also reported in the Geophysics community, specifically to applications of Full Waveform Inversion (FWI). Stochastic gradient descent competes with the L-BFGS algorithm, which is also widely used. Stochastic gradient descent has been used since at least 1960 for training linear regression models, originally under the name ADALINE. Another stochastic gradient descent algorithm is the least mean squares (LMS) adaptive filter.

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$\bar{w} = \frac{1}{t} \sum_{i=0}^{t-1} w_i$. When optimization is done, this averaged parameter vector takes the place of w .

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$G_{j,j} = \sum_{\tau=1}^t g_{\tau,j}^2$. This vector essentially stores a historical sum of gradient squares by dimension and is updated after every iteration. The formula for an update is now

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Implicit updates (ISGD) As mentioned earlier, classical stochastic gradient descent is generally sensitive to learning rate η . Fast convergence requires large learning rates but this may induce numerical instability. The problem can be largely solved by considering implicit updates whereby the stochastic gradient is evaluated at the next iterate rather than the current one:

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subject to additional stochastic noise. This approximation is only valid on a finite time-horizon in the following sense: assume that all the coefficients Q_i are sufficiently smooth. Let $T > 0$ and $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a sufficiently smooth test function. Then, there exists a constant $C > 0$ such that for all $\eta > 0$